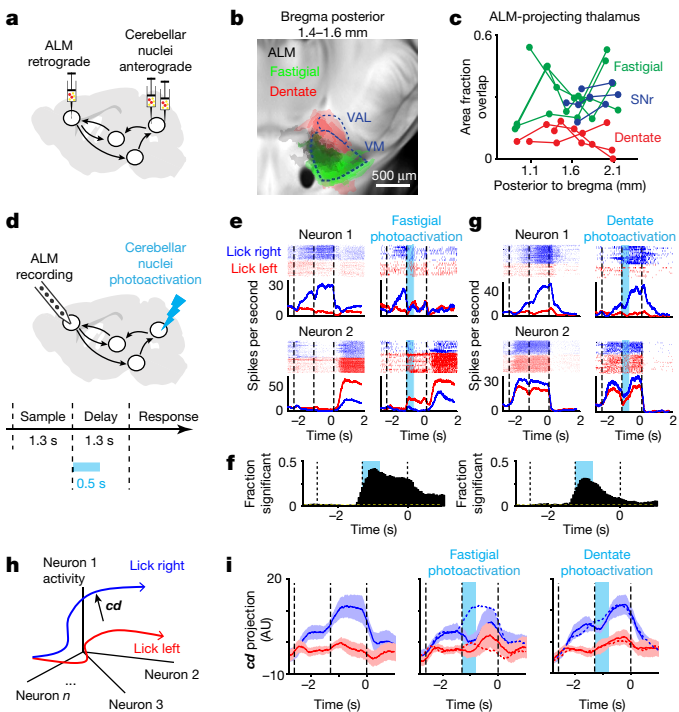




**Fig. 4 | The fastigial nucleus is critical for ALM coding of future movement.** **a**, Triple injections to map connectivity in the thalamus. **b**, Labelled areas in the thalamus from fastigial (green,  $n = 6$ ), dentate (red,  $n = 4$ ), and ALM (black,  $n = 8$ ) injections. Brain sections were aligned to the Allen Reference Brain (Methods). The ventral medial nucleus (VM) and parts of the ventral anterior-lateral nucleus (VAL) borders are based on the Allen Reference Atlas. Labelling was also present in the medial dorsal thalamus but is not quantified here. **c**, Overlaps between ALM-projecting thalamus and projections from the fastigial nucleus, dentate nucleus or basal ganglia substantia nigra reticulata (SNr). The co-labelled area is normalized to the total thalamic area labelled by each projection (area fraction overlap). Dots represent individual coronal sections; lines represent individual injections. **d**, ALM recording during contralateral fastigial or dentate photoactivation. Laser power, 0.5–1.5 mW. **e**, Spike raster plots (top) and spike rates (bottom) of two example ALM neurons. **f**, Fraction of neurons with significant change in spike rate compared to control trials ( $P < 0.01$ , two-tailed  $t$ -test);  $n = 446$ . The cyan bar indicates the photoactivation period. **g**, Same as **e** and **f**, but for dentate photoactivation;  $n = 466$ . **h**, Schematic of movement-specific activity trajectories and coding direction ( $cd$ ) in activity space. **i**, ALM activity in control and photoactivation trials projected onto the  $cd$ . Both correct and error trials are included, grouped by instructed movement. Mean  $\pm$  s.e.m. ( $n = 28$  sessions, 10 mice). The  $cd$  was estimated using an independent subset of control trials (correct trials only).